Module 04: Cognition in High School

Learning Objectives

1. Define **Cognition** within the context of High School.
2. Analyze how Cognition interacts with other components of the Active Inference framework.
3. Apply specific constraints of High School to the formal definition of Cognition.

Introduction

This module explores **Cognition**. In the **High School** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Cognition is a critical component of the 8-part Active Inference spine, bridging the gap between Perception and Action.

Key Concepts

1. Cognition as a Markov Blanket Boundary

How does Cognition define the boundary between the agent and the environment?

2. Generative Models of Cognition

What parameters involved in Cognition must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Cognition drive the perception-action loop?

Applications

In High School, we see Cognition manifest in: * **Specific Example 1**: The mathematical concept of entropy H = -sum(p*log(p)) directly quantifies cognitive uncertainty: when you are studying for a multiple-choice exam and have no idea which answer is correct, entropy is maximized (all options equally likely); as you study and your generative model improves, entropy decreases -- cognition is literally the mathematical process of reducing the entropy of your belief distribution. * Specific Example 2*: Matrix multiplication in neural network mathematics demonstrates cognitive inference: an input vector (sensory data) is multiplied by a weight matrix (the generative model's learned parameters) to produce an output prediction, and the difference between the prediction and reality (the error vector) is backpropagated to update the weights -- showing how cognition can be formalized as iterative matrix operations that minimize a loss function analogous to free energy.

Conclusion

Understanding Cognition allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.